

$$u'' + u = \tan x \text{ の解き方}$$

$$\partial = \frac{\partial}{\partial x} \text{ とおく,}$$

$$\boxed{\text{補題}} \quad (\partial^2 + 1)u = f(x) \text{ の解の1つを } u(x) = \int^x \sin(x-t)f(t)dt \text{ と作れる.}$$

$$\boxed{\text{証明}} \quad u(x) = \int^x \sin(x-t)f(t)dt \text{ とおく,}$$

$$u'(x) = \sin(x-x)f(x) + \int^x \cos(x-t)f(t)dt = \int^x \cos(x-t)f(t)dt,$$

$$u''(x) = \cos(x-x)f(x) - \int^x \sin(x-t)f(t)dt = f(x) - u(x),$$

$$u''(x) + u(x) = f(x),$$

q.e.d.

$$\boxed{\text{注意}} \quad v = u' \text{ とおく, } (\partial^2 + 1)u = f \text{ は } \frac{d}{dx} \begin{bmatrix} u \\ v \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix} \begin{bmatrix} u \\ v \end{bmatrix} + \begin{bmatrix} 0 \\ f \end{bmatrix} \text{ と書き直される.}$$

一般に $\frac{d}{dx} u(x) \stackrel{(*)}{=} Au(x) + b(x)$ の解の1つは以下のようにして作れる.

$$u(x) = e^{Ax} y(x) \text{ とおく, } \frac{d}{dx} u(x) = Au(x) + e^{Ax} \frac{d}{dx} y(x) \text{ なのて, } (*) \Leftrightarrow \frac{d}{dx} y(x) = e^{-Ax} b(x),$$

$$\text{ゆえに, } y(x) = \int^x e^{-At} b(t) dt, \quad u(x) = e^{Ax} y(x) = \int^x e^{A(x-t)} b(t) dt \text{ は } (*) \text{ の解になる.}$$

$$A = \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix} \text{ のとき, } e^{A(x-t)} = \cos(x-t)E + \sin(x-t)A = \begin{bmatrix} \cos(x-t) & \sin(x-t) \\ -\sin(x-t) & \cos(x-t) \end{bmatrix} \text{ なのて}$$

$$b(x) = \begin{bmatrix} 0 \\ f(x) \end{bmatrix} \text{ のとき, } e^{A(x-t)} b(t) = \begin{bmatrix} \sin(x-t)f(t) \\ \cos(x-t)f(t) \end{bmatrix} \text{ となり, 上の補題が得られる.} \quad \square$$

$$\text{上の補題より, } (\partial^2 + 1)u = \tan x \text{ の解の1つを } u(x) = \int^x \sin(x-t)\tan t dt \text{ と作れる.}$$

$$\sin(x-t)\tan t = (-\cos x \sin t + \sin x \cos t)\tan t = -\cos x \sin t \tan t + \sin x \sin t.$$

$$\int \sin t dt = -\cos t. \quad (\text{積分定数は略す.})$$

$$\int \sin t \tan t dt = \int \frac{1 - \cos^2 t}{\cos t} dt = \int \left(\frac{1}{\cos t} - \cos t \right) dt = \int \frac{dt}{\cos t} - \sin t.$$

$$s = \sin t \text{ とおく, } ds = \cos t dt$$

$$\int \frac{dt}{\cos t} = \int \frac{\cos t}{\cos^2 t} dt = \int \frac{\cos t}{1 - \sin^2 t} dt \stackrel{\downarrow}{=} \int \frac{ds}{1 - s^2}$$

$$= \frac{1}{2} \left(\frac{1}{1+s} + \frac{1}{1-s} \right) ds = \frac{1}{2} \left(\log(1+s) - \log(1-s) \right) = \frac{1}{2} \log \frac{1+s}{1-s} = \frac{1}{2} \log \frac{1+\sin t}{1-\sin t}.$$

$$\int \sin(x-t)\tan t dt = -\frac{1}{2} \cos x \log \frac{1+\sin t}{1-\sin t} + \cos x \sin t - \sin x \cos t$$

$$= -\frac{1}{2} \cos x \log \frac{1+\sin t}{1-\sin t} - \sin(x-t).$$

ゆえに, 次は $u''(x) + u(x) = \tan x$ の解の1つになっている!

$$u(x) = \int^x \sin(x-t)\tan t dt = -\frac{1}{2} \cos x \log \frac{1+\sin x}{1-\sin x},$$